

NAME

landlock_create_ruleset – create a new Landlock ruleset

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <linux/landlock.h> /* Definition of LANDLOCK_* constants */
#include <sys/syscall.h> /* Definition of SYS_* constants */

int syscall(SYS_landlock_create_ruleset,
            const struct landlock_ruleset_attr *attr,
            size_t size, uint32_t flags);
```

DESCRIPTION

A Landlock ruleset identifies a set of rules (i.e., actions on objects). This **landlock_create_ruleset()** system call enables creating a new file descriptor identifying a ruleset. This file descriptor can then be used by **landlock_add_rule(2)** and **landlock_restrict_self(2)**. See **landlock(7)** for a global overview.

attr specifies the properties of the new ruleset. It points to the following structure:

```
struct landlock_ruleset_attr {
    __u64 handled_access_fs;
};
```

handled_access_fs is a bitmask of actions that is handled by this ruleset and should then be forbidden if no rule explicitly allows them (see **Filesystem actions** in **landlock(7)**). This enables simply restricting ambient rights (e.g., global filesystem access) and is needed for compatibility reasons.

size must be specified as `sizeof(struct landlock_ruleset_attr)` for compatibility reasons.

flags must be 0 if *attr* is used. Otherwise, *flags* can be set to:

LANDLOCK_CREATE_RULESET_VERSION

If *attr* is NULL and *size* is 0, then the returned value is the highest supported Landlock ABI version (starting at 1). This version can be used for a best-effort security approach, which is encouraged when user space is not pinned to a specific kernel version. All features documented in these man pages are available with the version 1.

RETURN VALUE

On success, **landlock_create_ruleset()** returns a new Landlock ruleset file descriptor, or a Landlock ABI version, according to *flags*.

ERRORS

landlock_create_ruleset() can fail for the following reasons:

EOPNOTSUPP

Landlock is supported by the kernel but disabled at boot time.

EINVAL

Unknown *flags*, or unknown access, or too small *size*.

E2BIG

size is too big.

EFAULT

attr was not a valid address.

ENOMSG

Empty accesses (i.e., *attr->handled_access_fs* is 0).

STANDARDS

Linux.

HISTORY

Linux 5.13.

EXAMPLES

See **landlock(7)**.

SEE ALSO

landlock_add_rule(2), **landlock_restrict_self(2)**, **landlock(7)**